

Henry Hengyuan Zhao

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Singapore

EDUCATION

- **National University of Singapore** Jan 2022 - Expected Sept. 2026
Ph.D., Electrical Computer Engineering Singapore
- **Nanjing University of Posts and Telecommunications** 2016 - 2020
B.S., Communication Engineering China

EXPERIENCE

- **Microsoft Research GenAI Team** Aug 2025 - June 2026
Research Intern, LMM RL Post-Training; Reward Model Designs; Model Behavior Evaluation remote, Singapore
- **Sea AI Lab** June 2023 - March 2024
Research Intern, Multimodal Large Language Models: Data Generation and Training Paradigm Singapore
- **Alibaba DAMO Academy** Dec. 2021 - March 2023
Research Intern, Parameter-Efficient Fine-tuning for Large Vision Transformers Singapore
- **MIG, SenseTime Inc** June 2021 - October 2021
Research Intern China
- **Vision Technology (VIS), Baidu Inc** Dec. 2020 - June 2021
Research Intern China
- **Shenzhen Institutes of Advanced Technology (SIAT), CAS** Sep. 2019 - Dec. 2021
Research Intern China

PUBLICATIONS

- Xiangyu Zhao, **Henry Hengyuan Zhao**, Yiheng Wang, Wanghan Xu, Yuhao Zhou, Qinglong Cao, Zhiwang Zhou, LEI BAI, Wenlong Zhang, Xiao-Ming Wu, "Sci-PRM: A Tool Aware Process Reward Model for Scientific Reasoning Verification", **ACM KDD 2026**
- Jiahao Tang*, **Henry Hengyuan Zhao***[†], Lijian Wu, Yifei Tao, Dongxing Mao, Wanyang, Jingru Tan, Min Zeng, Min Li, Alex Jinpeng Wang, "From Charts to Code: A Hierarchical Benchmark for Multimodal Models", **ACL Main 2026**.
*: Equal first author. [†]: Project Lead.
A new challenging benchmark "Chart2Code", which spans three task levels and over 2,000 tasks across 22 chart types; ran large-scale experiments on 29 leading models.
- **Henry Hengyuan Zhao**, Kaiming Yang, Wendi Yu, Difei Gao, and Mike Zheng Shou, "WorldGUI: An Interactive Benchmark for Desktop GUI Automation from Any Starting Point", **ACL 2025 Workshop**.
Benchmark: An interactive evaluation framework for testing GUI agents in dynamic, stateful environments, with automated correctness verification at each step. Agent: A general agent framework built on a critic/self-verification loop, relevant to building AI agents with guardrails and automated correctness checking.
- **Henry Hengyuan Zhao**, Wenqi Pei, Yifei Tao, Mike Zheng Shou, "InterFeedback: Unveiling Interactive Intelligence of Large Multimodal Models via Human Feedback", **EMNLP 2025 Findings**.
Built an interactive evaluation framework and benchmark to study whether frontier multimodal models can improve through real-time human feedback loops. Key findings: (1) accuracy alone may not fully reflect model intelligence; (2) models may over-cater to human preferences rather than genuinely correct errors; (3) low-quality feedback can degrade performance more than simple binary (0/1) signals – directly informing how human-in-the-loop feedback should be designed for guiding/verifying AI systems.
- **Henry Hengyuan Zhao**, Pan Zhou, Difei Gao, Zechen Bai, Mike Zheng Shou, "LOVA³: Learning to Visual Question Answering, Asking and Assessment", **NeurIPS, 2024**.
Only answering questions? Let's think about asking and assessing questions when training MLLMs? Without hyperparameter tuning or additional data annotation, consistent performance improvements are achieved!
- **Henry Hengyuan Zhao**, Pan Zhou, Mike Zheng Shou, "Genixer: Empowering Multimodal Large Language Model as a Powerful Data Generator", **ECCV, 2024**.
How MLLMs perform in data generation? Proposed the first framework turning MLLMs into scalable data generators, producing 1M samples.

- **Henry Hengyuan Zhao**, Pichao Wang, Yuyang Zhao, Hao Luo, Fan Wang, Mike Zheng Shou, "SCT: A Simple Baseline for Parameter-Efficient Fine-Tuning via Salient Channels", *IJCV*, 2023.
We found that tuning only a small number of task-specific channels, referred to as salient channels, is sufficient. This work represents a remarkable reduction of $780 \times$ in parameter costs compared to its full fine-tuning counterpart.
- **Henry Hengyuan Zhao**, Hao Luo, Yuyang Zhao, Pichao Wang, Fan Wang, Mike Zheng Shou, "Revisit Parameter-Efficient Transfer Learning: A Two-Stage Paradigm", *Arxiv*, 2023.
- Yihao Liu, **Hengyuan Zhao**, Jinjin Gu, Yu Qiao, Chao Dong, "Evaluating the Generalization Ability of Super-resolution Networks", *TPAMI*, 2023.
- Yihao Liu*, **Hengyuan Zhao***, Kelvin CK Chan, Xintao Wang, Chen Change Loy, Yu Qiao and Chao Dong, "Temporally Consistent Video Colorization with Deep Feature Propagation and Self-regularization Learning", *CVM*, 2023.
- Xiangtao Kong, **Hengyuan Zhao**, Qiao Yu and Chao Dong, "ClassSR: A General Framework to Accelerate Super-Resolution Networks by Data Characteristic", *IEEE Conference on Computer Vision and Pattern Recognition, CVPR*, 2021.
- **Hengyuan Zhao**, Xiangtao Kong, Jingwen He, Yu Qiao and Chao Dong, "Efficient Image Super-Resolution using Pixel Attention", *European Conference on Computer Vision Workshop (ECCV Workshop)*, 2020).
- **Hengyuan Zhao**, Wenze Shao, Bingkun Bao and Haibo Li, "A Simple and Robust Deep Convolutional Approach to Blind Image Denoising", *International Conference on Computer Vision Workshop (ICCV Workshop)*, 2019).
- **Hengyuan Zhao***, Wenhao Wu*, Yihao Liu*, Dongliang He, "Color2Embed: Fast Exemplar-Based Image Colorization using Color Embeddings", *Arxiv*, 2021.

RESEARCH SUPERVISION AND MENTORSHIP

- **Henry Hengyuan Zhao (Project Lead)**, "From Charts to Code: A Hierarchical Benchmark for Multimodal Models", *ACL* 2026.
Mentored one Master's student and one undergraduate student. Conceived the research idea and overall benchmark structure for agentic coding. Led literature review, task taxonomy design, data collection and annotation protocols, evaluation metrics, and large-scale experiments. Directed paper writing and positioning relative to existing multimodal and agentic benchmarks.
- **Henry Hengyuan Zhao (Project Lead)[†]**, "ChartJudge: Evaluating Reward Models for LMM-based Chart-to-Code Generation", *AAAI*, Under Review.
Mentored one undergraduate student. Initiated and led the development of a new benchmark for scientific chart judgment, targeting reward model design for vision-language models. Guided problem definition, dataset construction, and early experimental analysis.

TEACHING ASSISTANTSHIP AND LAB SUPERVISION

- **Teaching Assistant and Lab Supervisor**, *CG2023 Signals and Systems*, National University of Singapore, Semester 2, 2022–2023.
Led tutorial sessions and laboratory supervision, providing hands-on support for in-lab practical assignments by answering questions and troubleshooting circuit design and software-related issues, while reinforcing core signal processing concepts.
- **Teaching Assistant**, *EE5112 Human-Robot Interaction (Graduate Level)*, National University of Singapore, Semester 1, 2023–2024. Assisted in designing one course projects. Mentored graduate students on advanced topics in human-centered AI and interaction-aware learning systems. Provided guidance on project scoping, literature review, and experimental evaluation.

SERVICE CONTRIBUTIONS

Journal Reviewer: International Journal of Computer Vision (IJCV), The IEEE Transactions on Image Processing (TIP).
Conference Reviewer: NeurIPS 2026, ICML 2026, ICLR 2026, ECCV 2026, ACM 2026, NeurIPS 2025, ACM Multimedia 2025, ACL 2025, WACV 2024, 2025, AAAI 2025
Regular reviewer for top-tier venues in machine learning, computer vision, and natural language processing.

TALKS

- I gave an Oral Presentation in the ICLR 2025 Bi-Align Workshop.
- I gave a talk about "Memory Efficient Techniques" at Show Lab internal meeting.
- I gave a talk on AIM 2020 Efficient Super-Resolution Challenges.